

Predicting and Explaining Employee Absenteeism at Work

Jennifer Huang, Shih-Hui Huang, Tse-Chun Hsu, Yu-Chen Lu

I. Project Idea

The goal of this project is to analyze the factors that influence employee absenteeism at work and to build models that can predict absenteeism time in hours. Employee absenteeism is an important workforce issue because frequent or prolonged absences can disrupt operations, reduce productivity, and increase labor costs. Using a real-world absenteeism dataset from a courier company in Brazil, this project will examine how demographic, lifestyle, transportation, work-related, and medical factors are associated with absence duration.

This project is also motivated by prior research that applied a neuro-fuzzy network to the same dataset to predict absenteeism. Building on that idea, this project will revisit the problem using interpretable exploratory analysis and machine learning methods. In addition to prediction, the project aims to uncover insights, such as which reasons for absence are linked to longer absence hours and whether factors like age, BMI, distance from work, education, smoking, drinking, or workload are associated with absenteeism.

II. Dataset

The dataset we plan to use is the [Absenteeism at Work](#) dataset, which contains records of absenteeism from July 2007 to July 2010 at a courier company in Brazil, downloaded from the UC Irvine Machine Learning Repository. Since this is an existing public dataset, no new data collection is required; instead, the data will be downloaded directly from the UCI repository and imported into Python for cleaning, exploration, and modeling.

The dataset contains 740 records and 21 attributes, with Absenteeism time in hours as the target variable. The features cover absence reasons, time-related factors, transportation, work history, age, workload, education, family information, health-related measures, and lifestyle habits such as drinking and smoking.

This dataset is a strong fit for the project because it aligns closely with the goal of understanding and predicting employee absenteeism. It is manageable for a course project, supports both exploratory analysis and machine learning, and has also been used in prior academic research, making it useful for method comparison and interpretable business insights.

III. Project Plan

The project will be implemented using Python and Jupyter Notebook for data cleaning, data visualization, and machine learning. Since the dataset is structured tabular data, the main focus will be on preprocessing categorical and numerical variables, conducting exploratory data analysis, and building models to predict absenteeism time.

We will begin with data cleaning and preprocessing, including checking for missing values, outliers, duplicate records, and proper encoding of categorical variables. Exploratory analyses will be conducted to examine the distribution of absenteeism hours and identify patterns across different factors and indicators. We also plan to utilize different machine learning models and compare their performance. The final goal is to identify both the most accurate model and the most important factors associated with absenteeism, so that the project provides not only prediction results but also meaningful business insights.

IV. Reference Articles/Paper

Llamas Blázquez, Pablo. "Predicting workplace absenteeism using machine learning: a pilot study in occupational health." *Journal of occupational medicine and toxicology (London, England)* vol. 20,1 38. 10 Nov. 2025, doi:10.1186/s12995-025-00482-5

Martiniano, Andréa et al. "Application of a neuro fuzzy network in prediction of absenteeism at work." *7th Iberian Conference on Information Systems and Technologies (CISTI 2012)* (2012): 1-4.

Skorikov, Mikhail et al. "Prediction of Absenteeism at Work using Data Mining Techniques." *2020 5th International Conference on Information Technology Research (ICITR)* (2020): 1-6.

Wahid, Zaman et al. "Predicting Absenteeism at Work Using Tree-Based Learners." *Proceedings of the 3rd International Conference on Machine Learning and Soft Computing* (2019): n. pag.

V. Work Division

- Jennifer Huang: topic ideation, project proposal, clustering
- Shih-Hui Huang: topic ideation, data preprocessing
- Tse-Chun Hsu: topic ideation, predictive models
- Yu-Chen Lu: topic ideation, exploratory data analysis